Complex Engineering Problem

Signal and Sytems

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December, 22nd, 2023

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Problem Statement

Design an active low-pass Sallen-Key 4th order filter, where precision and functionality converge. Design the filter to achieve a cutoff frequency of precisely ***1 kHz***, ensuring it strikes the right balance between allowing desired signals and suppressing unwanted noise. Set the voltage gain at ***10 dB*** to amplify signals with finesse. Utilize the proven Sallen-Key topology to construct a 4th order filter, a widely recognized method in analog circuit design. Navigate the standard E12 resistor values and capacitor values for consistency and practicality. Employ an ideal operational amplifier with a unity gain bandwidth exceeding **10 times the cutoff frequency**. Power the operational amplifier with a single ***+15V*** supply for simplicity. Ensure that the input and output impedances of the filter harmonize within practical limits. Optimize the filter for a Butterworth response, emphasizing a maximally flat passband.

Your challenge is to synthesize these elements, producing not just a filter but a thoughtfully engineered solution

Abstract

This report presents a meticulously engineered solution for an active low-pass Sallen-Key 4th order filter, harmonizing precision and functionality in analog circuit design. The filter is tailored for a cutoff frequency precisely set at 1 kHz, striking a balance between accommodating desired signals and suppressing unwanted noise. Leveraging the well-established Sallen-Key topology, renowned for its simplicity and efficacy, the design navigates standard E12 resistor and capacitor values for consistency and practicality. With a targeted voltage gain of 10 dB, the filter optimally balances signal amplification with finesse.

The choice of an ideal operational amplifier, featuring a unity gain bandwidth exceeding 10 times the cutoff frequency, ensures signal integrity. Operating on a single +15V supply streamlines the design without compromising performance. The filter's response is optimized for a Butterworth characteristic, emphasizing a maximally flat passband. Component values are meticulously selected to align with standard industry practices.

In the pursuit of harmony, input and output impedances are finely tuned within practical limits. This paper not only presents a filter design but encapsulates a thoughtfully engineered solution that seamlessly integrates precision, functionality, and practicality into the landscape of analog signal processing.

Sallen-Key Active High Low Pass Filter

The Sallen-Key topology is a popular and widely used circuit configuration in active filter design, known for its simplicity and effectiveness, particularly in realizing low-pass and high-pass filters. The Sallen-Key filter is typically implemented using operational amplifiers and passive components such as resistors and capacitors. The working principle of a Sallen-Key filter can be explained in the context of a second-order low-pass filter

1. Basic Configuration:
The Sallen-Key filter consists of two resistor-capacitor (RC) networks connected to the inverting (-) and non-inverting (+) inputs of an operational amplifier (op-amp).

2. Feedback Network:
The feedback network typically includes two resistors (R1 and R2) and a capacitor (C1). These components create a voltage divider network that provides feedback to the inverting input of the op-amp.

3. Inverting Input:
The inverting input receives the feedback voltage from the resistor-capacitor network. The feedback network, along with the op-amp's characteristics, determines the filter's transfer function.

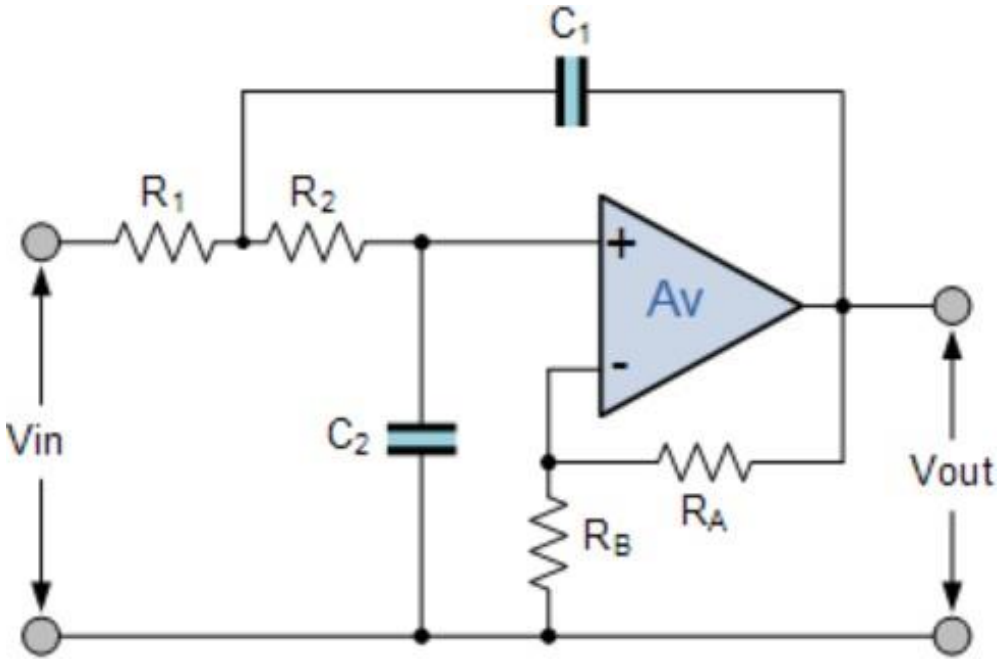
4. Frequency Response:
The values of resistors and capacitors in the feedback network determine the cutoff frequency of the filter. At frequencies below the cutoff, the filter provides significant signal attenuation, acting as a low-pass filter.

5. Op-Amp Operation:
The op-amp amplifies the difference between its inverting and non-inverting inputs. The feedback network influences the op-amp's behavior to achieve the desired filtering characteristics.

6. Order of the Filter:
The Sallen-Key topology can be cascaded to achieve higher-order filters by connecting multiple filter stages in series.

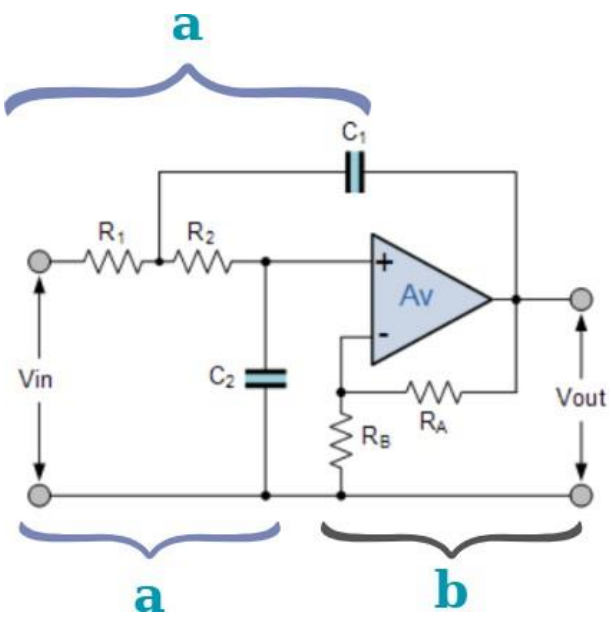
To implement a fourth-order filter, two Sallen-Key biquad stages are cascaded, connecting the output of the first stage to the input of the second. This arrangement effectively combines the transfer functions of two second-order filters, contributing four poles to the overall response. The component values of resistors and capacitors in each stage determine the filter's characteristics. Careful consideration of component values is crucial to maintain stability and prevent issues such as peaking.
The fourth-order Sallen-Key biquad configuration is valuable in applications requiring a steeper roll-off and precise frequency control. Its simplicity, efficiency, and customizable response make it a practical choice for designers seeking enhanced filtering capabilities in compact and economical circuit implementations.

Basic Configuration:



Calculations

To understand this topology better, we divide this circuit into two parts *a) filtering part* , which includes C1, C2, R1, R2 and *b) the gain part*, which includes the opamp, Rb, Ra.



Calculations for a.

$$f_c = \frac{1}{2\pi RC}$$

where either C or R is assumed and used to calculate the other component's value

Calculations for b.

$$gain,\ A_v = 1 + \frac{R_a}{R_b}$$

where either Ra or Rb is assumed and used to calculate the other component's value

Assuming,

$$R_1 = R_2 = R_b = R$$

$$C_1 = C_2 = 0.1uF\ ^{1}$$

$$f_c = \frac{1}{2\pi RC} \text{ where } f_c = 1kHz$$

Calculating value for Resistance

$$R = \frac{1}{2\pi C f_c}$$

$$R = \frac{1}{2\pi * 0.1uF * 1kHz} = 1.59k\Omega$$

The gain provided in the problem statement is **10dB**, so first we will convert it into a voltage value. The formula is given by :

$$A_v = 10^{\ db/20}$$

[1] Capacitance Value chosen as 0.1uF due to its availability

Where 20db is the roll off value of a low pass filter, which is fixed.

$$A_v = 10^{5/20} = 1.778V$$

Here we use 5 as the dB value since we are using 2 biquads and each will have 5db and after cascading it becomes 10db.

Calculating R_a

$$A_v = 1 + \frac{R_a}{R_b}$$

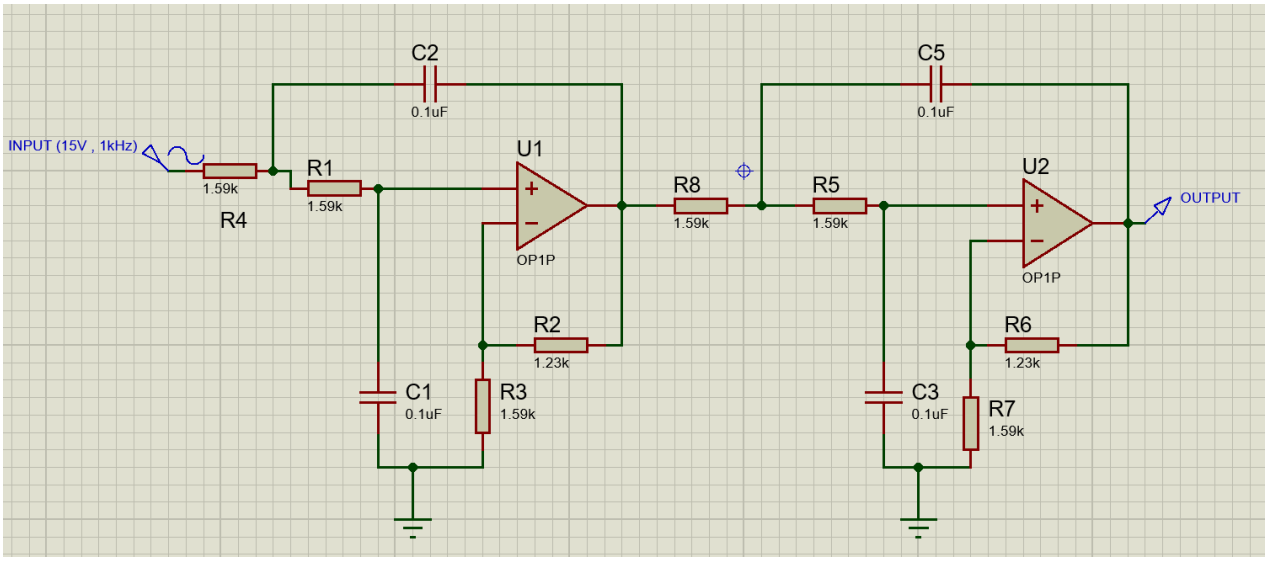
$$R_a = R_b * (A_v - 1)$$

$$R_a = 1.59k\Omega * (1.778 - 1)$$

$$R_a = 1.237k\Omega$$

Sallen-Key 4th Order Filter (ideal op-amp)

Configuration with 2 Sallen-Keys Topologies Cascaded Together

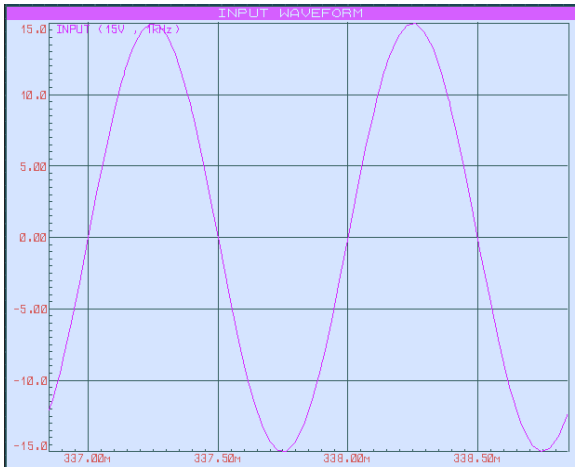


Components used:

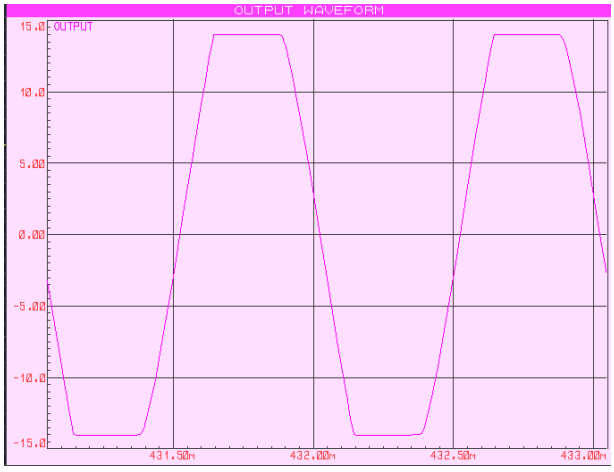
1.59kΩ Resistors = 6 0.1µF Capacitors = 4 Ideal Op-Amp = 2

1.23kΩ Resistors = 2

Input wave Signal

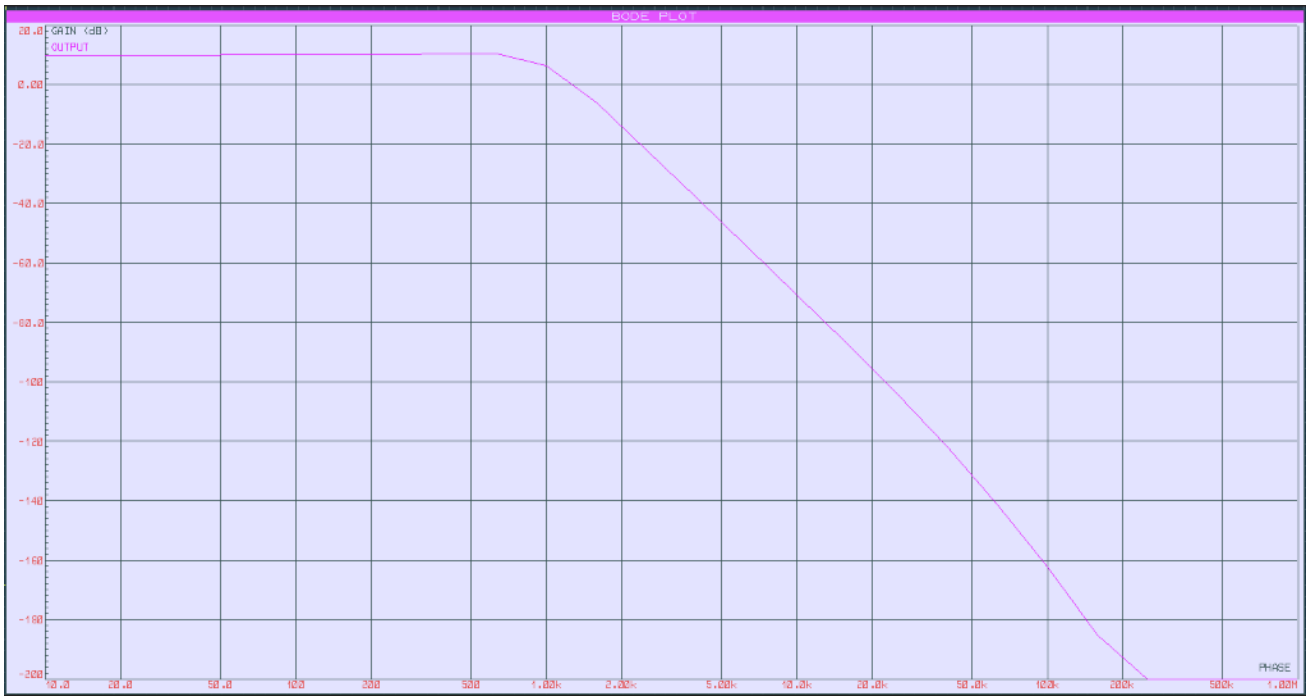


Output Wave Signal



The Bode plot is a graphical representation that depicts the frequency response of a system.

In this specific case, the gain is at 10dB, which means that the system amplifies the input signal by a factor of 10 at a particular frequency. The cutoff frequency, which is 1kHz in this instance, signifies the point at which the gain starts decreasing.

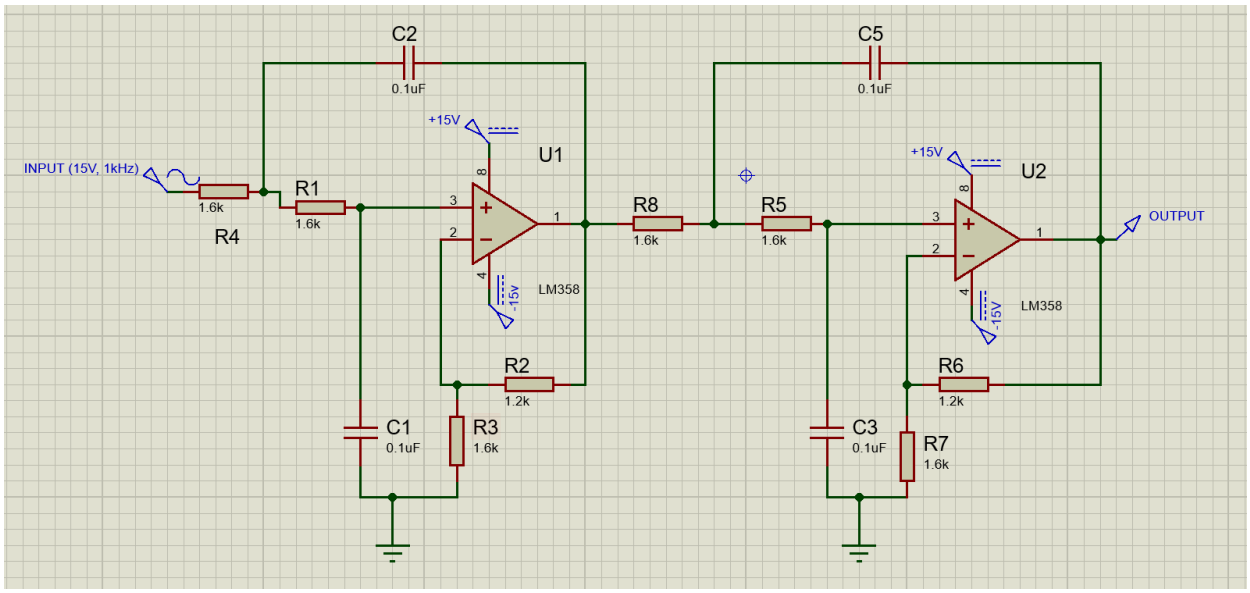


To break it down further, at frequencies below 1kHz, the system exhibits a gain of 10dB or more, suggesting amplification. As the frequency surpasses 1kHz, the gain starts to decrease, indicating that the system becomes less effective in amplifying signals at higher frequencies. The cutoff frequency is essentially the point where the system response begins to attenuate or reduce the signal strength.

The response above was exhibited by an ideal operational amplifier operating in ideal conditions i.e. exact resistor values, exact capacitor values and an opamp with unity gain bandwidth.

Sallen-Key 4th Order Filter (LM-358)

Configuration with 2 Sallen-Keys Topologies Cascaded Together

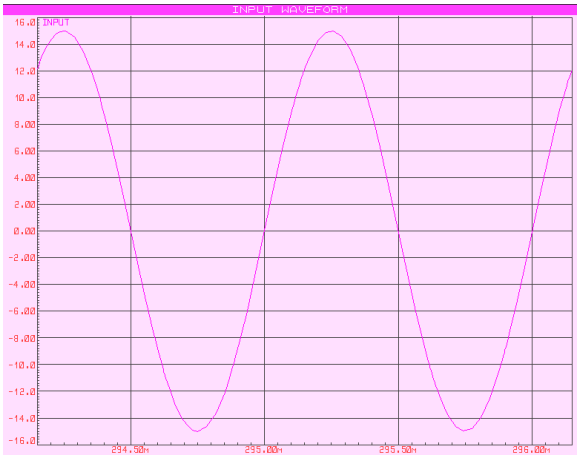


Components used:

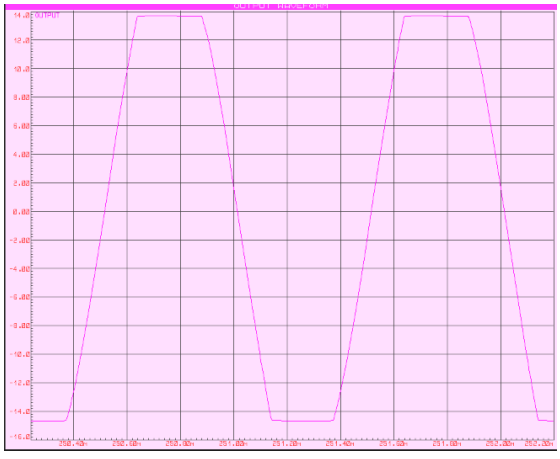
1.6kΩ Resistors = 6 0.1µF Capacitors = 4 LM358=2

1.20kΩ Resistors = 2

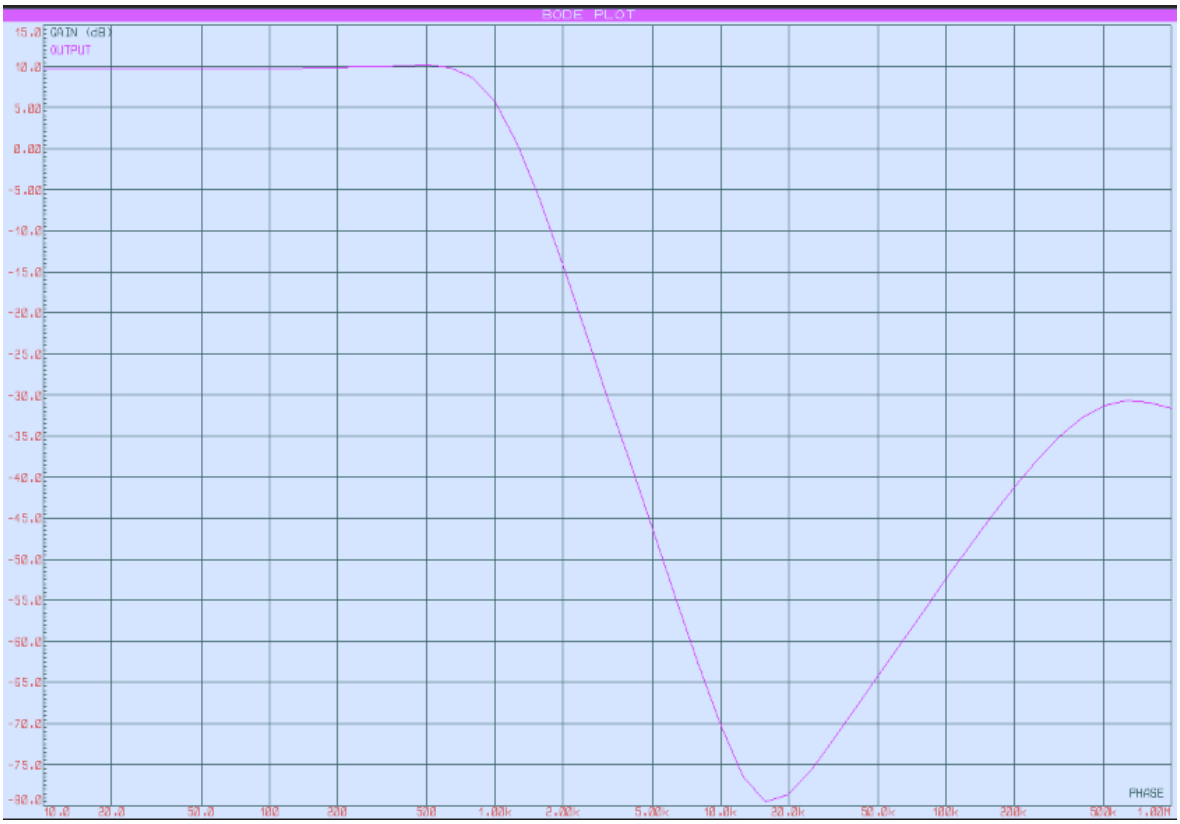
Input wave Signal



Output wave signal

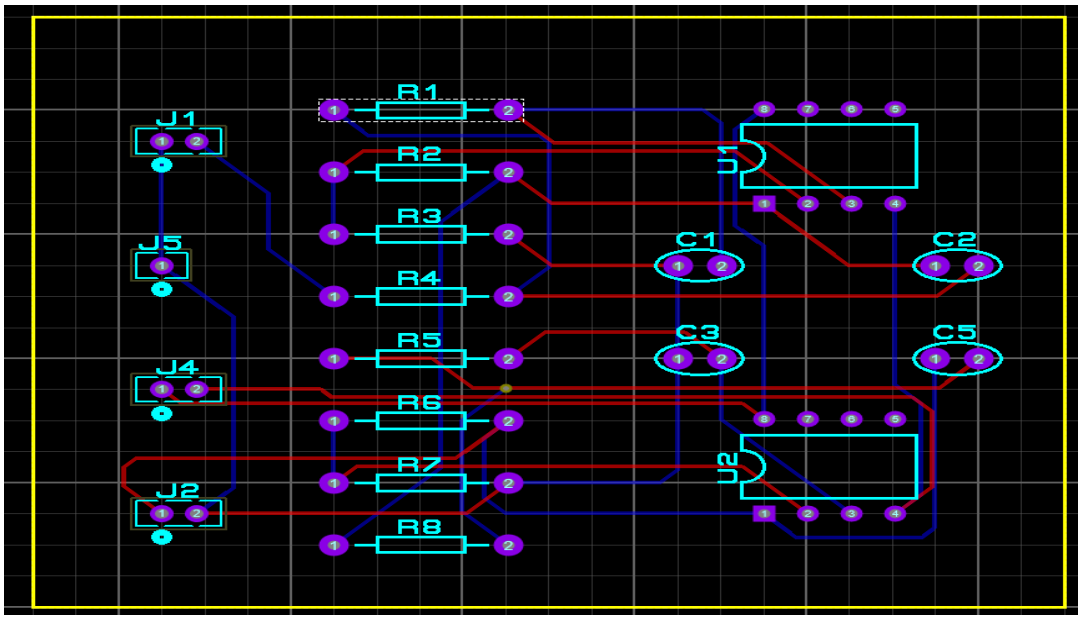


When employing a real-world operational amplifier (op-amp) rather than an idealized counterpart, deviations from the expected frequency response may occur due to several non-ideal characteristics inherent in practical op-amp designs. Finite bandwidth, stemming from the op-amp's inability to amplify signals at extremely high frequencies, plays a significant role in shaping the response.

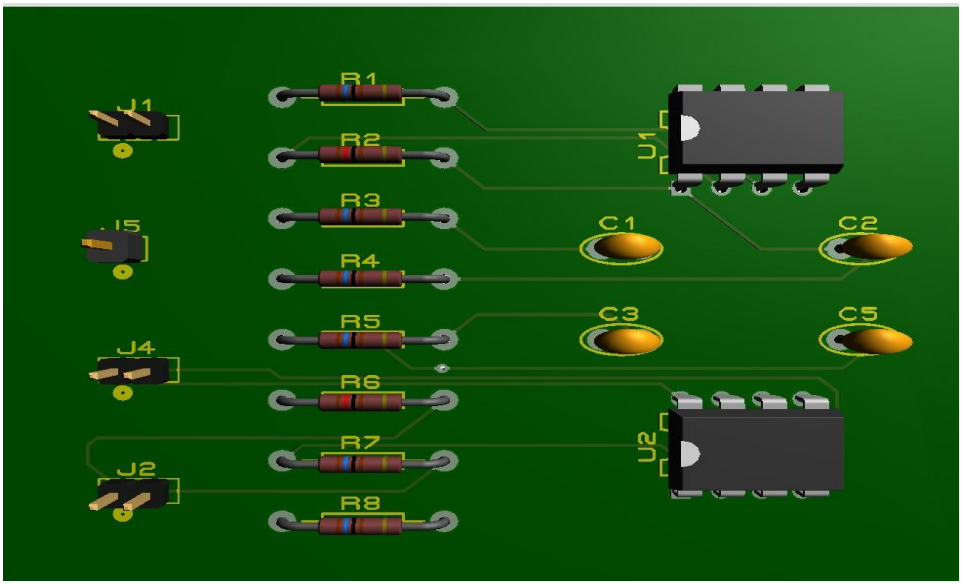


When a rise in gain is observed beyond the cutoff frequency, it reflects the intricate interplay of these non-ideal factors, emphasizing the importance of careful consideration in op-amp selection and circuit design to manage and minimize such deviations.

PCB Design



The pcb design of the filter, with real capacitor, resistor values and op-amp model



Conclusion

In conclusion, employing the Sallen-Key topology with an ideal operational amplifier and the LM358 offers a practical yet effective approach in analog filter design. The LM358, chosen for its cost-effectiveness and widespread availability, introduces real-world considerations such as finite input and output impedance, bandwidth limitations, and a specified common-mode voltage range.

Despite these deviations from ideal op-amp characteristics, the LM358 provides a practical solution for various applications, particularly when factors like simplicity and affordability are paramount.

These non-idealities, ensuring proper compensation and practical adjustments to optimize the filter's performance must be kept in mind when designing a filter.

By striking a balance between cost-effectiveness and functionality and leveraging the LM358's versatility, engineers can successfully implement Sallen-Key filters tailored to their specific application requirements.

References

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Sallen-key Low Pass Filters by Cadence System Analysis
Sallen and Key filters by Electronic Tutorial